

Notes: Grullon, Underwood, and Weston (JFE, 2014)

Comovement and Investment Banking Networks

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1 Big picture

- **Question.** Do investment banking relationships create investor networks that affect stock prices and trading behavior?
- **Core idea.** Investment banks repeatedly connect corporate issuers with investor clienteles. If these clienteles trade and hold stocks in segmented groups, firms that use the same lead underwriter should comove more with each other.
- **Main setting.** The paper studies US IPOs and SEOs over 1980–2008. For each offering, the authors form a value-weighted portfolio of firms associated with the same lead underwriter.
- **Main IPO result.** After an IPO, the IPO firm’s stock comoves more with stocks associated with the same underwriter than with stocks associated with nearby underwriters in the league-table ranking.
- **Institutional ownership channel.** Institutions associated with the lead underwriter hold larger stakes in the IPO firm than institutions associated with nearby banks, and this pattern persists over time.
- **SEO switch test.** When a firm switches lead underwriters between offerings, its stock comoves more with the new bank’s network and less with the old bank’s network.
- **Identification logic.** The SEO switch design uses the same firm before and after an offering, reducing concerns that fixed firm characteristics explain the network effect.
- **Mechanism evidence.** Changes in comovement are related to changes in institutional ownership by the new bank’s client institutions.
- **Trading evidence.** For Nasdaq firms, market-making activity moves from the old bank to the new bank around the SEO, supporting the view that the network changes who trades the stock.
- **Robustness.** The results hold in matched samples, in forced-switch samples caused by bank mergers and closures, after considering risk factors, and under alternative event windows.
- **Economic takeaway.** Underwriting relationships do not merely certify firms or distribute securities. They create segmented investor networks that can affect secondary-market prices, ownership, and trading behavior.

2 Data and measurement

2.1 Investment banking networks

- A firm is associated with a bank if the bank was the sole lead underwriter for the firm’s IPO or SEO within the last two years.
- For each focal firm, the *lead bank portfolio* is the value-weighted portfolio of all other firms associated with the same lead underwriter.
- The control portfolios are associated with banks ranked directly above and below the lead bank in underwriting volume during the offering year.
- The focus on adjacent league-table banks creates a comparison group with similar reputation and scale.

Interpretation. The empirical object is not a formal social graph but a bank-specific clientele network inferred from repeated underwriting relationships.

2.2 IPO sample

- The IPO sample contains 2,540 offerings from 1980–2008.

- The firm must use a single lead underwriter ranked among the top 25 by underwriting volume in the offering year.
- The offering must be at least \$5 million.
- The firm must have at least one year before its first SEO, so post-IPO comovement can be measured cleanly.

Interpretation. The IPO analysis asks whether the initial underwriter creates a new network affiliation that shows up in returns and ownership immediately after the offering.

2.3 SEO sample

- The SEO sample contains 2,869 offerings.
- Each SEO uses a single lead underwriter, as did the firm's most recent prior equity offering.
- 1,511 SEOs use the same lead bank as the previous offering, while 1,358 switch lead banks.
- The lead bank in the current offering must be ranked among the top 25.
- The analysis excludes the week before and after the SEO to avoid contamination from offering-window effects.

Interpretation. The SEO sample is the central identification setting because it observes the same firm before and after a change in underwriting network.

2.4 Institutional ownership measures

- An institution is associated with a bank if it ranks above the 75th percentile in holdings of firms associated with that bank.
- The paper studies both the fraction of the firm held by bank-associated institutions and the number of bank-associated institutions holding shares.
- For SEO switches, the paper separately tracks institutions associated with the old bank and the new bank.

Interpretation. Ownership changes help distinguish a pure return-correlation pattern from a network mechanism in which investor clientele changes after the offering.

2.5 Market-making and turnover measures

- For Nasdaq-listed firms, the paper measures dealer market share from Nasdaq monthly volume share reports.
- The market-making analysis compares the old underwriter and the new underwriter for firms that switch banks.
- The paper also studies turnover comovement: whether turnover in the focal stock begins to comove more with turnover of stocks in the new bank network.

Interpretation. These tests ask whether the same network migration appears in trading activity, not only in returns and ownership.

3 Models and empirical specifications

3.1 IPO comovement regression

For each IPO firm i , the baseline one-bank regression is

$$R_{i,t} = \alpha_i + \beta_{\text{Lead},i} R_{\text{Lead},t} + \epsilon_{i,t}. \quad (1)$$

The control-bank regressions are

$$R_{i,t} = \alpha_i + \beta_{\text{Bank1},i} R_{\text{Bank1},t} + \epsilon_{i,t}, \quad (2)$$

$$R_{i,t} = \alpha_i + \beta_{\text{Bank2},i} R_{\text{Bank2},t} + \epsilon_{i,t}. \quad (3)$$

The multivariate regression is

$$R_{i,t} = \alpha_i + \beta_{\text{Lead},i} R_{\text{Lead},t} + \beta_{\text{Bank1},i} R_{\text{Bank1},t} + \beta_{\text{Bank2},i} R_{\text{Bank2},t} + \epsilon_{i,t}. \quad (4)$$

Interpretation. The lead-bank beta and R^2 measure how much the IPO stock moves with the network associated with its own underwriter, compared with nearby bank networks.

3.2 SEO same-bank pre/post regression

Around SEOs, the paper estimates

$$R_{i,t} = \alpha_i + \beta_{\text{Lead},i}^{\text{Pre}} R_{\text{Lead},t} + \epsilon_{i,t} \quad (5)$$

for the pre-SEO window and

$$R_{i,t} = \alpha_i + \beta_{\text{Lead},i}^{\text{Post}} R_{\text{Lead},t} + \epsilon_{i,t} \quad (6)$$

for the post-SEO window. The main object is

$$\Delta \beta_{\text{Lead},i} = \beta_{\text{Lead},i}^{\text{Post}} - \beta_{\text{Lead},i}^{\text{Pre}}. \quad (7)$$

Interpretation. For firms that do not switch banks, this captures strengthening of the existing network relation. For firms that switch banks, it captures movement into the new bank's network.

3.3 SEO old-bank/new-bank regression

For switchers, the bivariate specification is

$$R_{i,t} = \alpha_i + \beta_{\text{New},i} R_{\text{New},i,t} + \beta_{\text{Old},i} R_{\text{Old},i,t} + \epsilon_{i,t}. \quad (8)$$

The key comparison is

$$(\beta_{\text{New},i}^{\text{Post}} - \beta_{\text{New},i}^{\text{Pre}}) - (\beta_{\text{Old},i}^{\text{Post}} - \beta_{\text{Old},i}^{\text{Pre}}). \quad (9)$$

Interpretation. If the bank network channel is real, comovement should increase with the new bank portfolio but not with the old bank portfolio.

3.4 Institutional ownership mechanism regression

The paper relates changes in comovement to changes in institutional ownership:

$$\Delta \text{Comovement}_i = \gamma_0 + \gamma_1 \text{DifferentBank}_i + \gamma_2 \text{FirstSEO}_i + \gamma_3 \text{RelativeSize}_i + \gamma_4 \text{YearsSinceLast}_i + \gamma_5 \text{OwnershipChange}_i + \epsilon_i \quad (10)$$

Interpretation. A positive coefficient on ownership change supports the idea that new bank institutions help produce the comovement change.

3.5 Turnover comovement regression

The turnover analogue is

$$\Delta TO_{i,t} = \alpha_i + \beta_{\text{Lead},i} \Delta TO_{\text{Lead},t} + \epsilon_{i,t}, \quad (11)$$

where $\Delta TO_{i,t}$ is the percentage change in turnover for stock i .

Interpretation. If investor networks shape trading, the turnover of a switcher’s stock should begin to move with turnover of the new bank network.

4 Identification and empirical design

4.1 Main identifying variation

- The IPO design compares same-bank comovement to adjacent-bank comovement after issuance.
- The SEO design compares the same firm before and after a change in lead underwriter.
- The old-bank/new-bank design asks whether the same firm shifts from one network to another.

Interpretation. The strongest evidence is not simply that same-bank firms comove. It is that switching underwriters changes which network the firm comoves with.

4.2 Why adjacent banks are useful controls

- Banks directly above and below the lead bank in underwriting rankings are similar in size and reputation.
- If comovement were driven only by underwriter quality or market-wide risk, the adjacent-bank portfolios should look similar.
- The lead-bank portfolio has larger beta and R^2 than the adjacent-bank portfolios.

Interpretation. The controls reduce the concern that any large investment bank’s client firms naturally comove for non-network reasons.

4.3 Why the switcher design is powerful

- A firm that switches from old bank to new bank is observed on both sides of the same event.
- Firm fixed characteristics are less likely to explain a shift from old-bank comovement to new-bank comovement.
- The paper separately examines firms forced to switch by bank mergers or closures, where the switch is less likely to reflect endogenous firm choice.

Interpretation. The forced-switch and matched-sample tests strengthen the causal interpretation, even though the paper is not a randomized experiment.

4.4 What the paper cannot fully prove

- Underwriter choice is not randomly assigned.
- Switching banks may coincide with changes in firm strategy, risk, analyst coverage, or investor demand.
- The network definition is based on underwriting relationships and institutional holdings, not direct investor communication data.
- The bond-market robustness result is described but not shown in a table.

Interpretation. The paper's case is strongest because returns, ownership, market making, and turnover all point toward the same network mechanism.

5 Tables and figures

The figures and tables below are directly cropped from the original paper and grouped to save space while preserving the paper’s visual evidence.

5.1 Table 1 + Table 2: sample construction and IPO comovement

Read: Table 1:

- The IPO sample contains 2,540 offerings; the SEO sample contains 2,869 offerings.
- In the SEO sample, 1,511 firms do not switch banks and 1,358 switch banks.
- Switchers wait longer since the last offering: 3.87 years versus 1.89 years for non-switchers.
- Average IPO offering size is about \$70.7 million; average SEO offering sizes are about \$91.8 million for non-switchers and \$89.5 million for switchers.

Table 1

Summary statistics.
This table presents summary statistics for the two main samples used in the paper. The initial sample consists of 2,540 initial public offerings (IPOs) over the period 1980–2008. We require that the firm use a single lead underwriter for the offering that was ranked in the top 25 banks by underwriting volume during the year of the offering. Only IPOs with at least one year until the first seasoned equity offering (SEO) are included. The second sample consists of 2,869 SEOs that used a single lead underwriter for both the SEO and their most recent equity offering and for which the lead bank on the current offering is ranked in the top 25. Only SEOs with at least six months after the most recent offering and with at least six months until the next offering are included in the sample. All IPOs and SEOs must be at least \$5 million in size. Offering Size is the amount (in millions) raised in the offering. #Firms in Bank Portfolio is the average number of firms in the portfolio associated with the bank leading the offering. For IPOs, #Firms in Bank 1,2,3 Portfolio is the average number of firms in the portfolio associated with the bank ranked directly above (below) the bank leading the IPO in terms of underwriting volume in the year of the IPO. Total # Institutions is the number of institutions holding a stake in the firm immediately following the IPO or SEO. For SEOs, Years Since Last Offering is the number of years since the firm’s last equity offering, and Relative Offering Size is offering size divided by the market capitalization of the firm in the month prior to the SEO.

	Mean	25th percentile	Median	75th percentile
Panel A: IPO Sample (2,540 offerings)				
Offering size (millions of dollars)	70.66	24.80	40.60	70.40
#Firms in bank portfolio	49.22	28.99	46.46	68.38
# Firms in bank 1 portfolio	48.48	27.27	45.76	69.62
# Firms in bank 2 portfolio	41.98	22.03	37.78	59.38
Total # institutions	19.41	5.00	16.00	28.00
Panel B: SEO Sample (2,869 offerings)				
Did not switch investment bank (1,511 offerings)	91.81	31.20	57.00	103.50
Offering size (millions of dollars)	0.16	0.09	0.13	0.20
Relative offering size	1.89	0.83	1.28	2.27
Years since last offering	46.61	27.68	43.87	61.47
# of firms in bank portfolio	75.13	35.00	57.00	92.00
Total # institutions				
Switched investment banks (1,358 offerings)				
Offering size (millions of dollars)	89.51	29.40	54.60	100.00
Relative offering size	0.18	0.09	0.15	0.23
Years since last offering	3.87	1.45	2.69	4.97
# of firms in bank portfolio	42.24	20.39	36.92	61.93
Total # institutions	30.69	34.00	58.00	101.00

Table 2

Comovement with other stocks associated with same bank following the initial public offering (IPO).
This table reports the average adjusted R^2 of the following regression models:

$$R_{i,t} = \alpha + \beta_{lead} R_{lead,t} + \epsilon_{i,t}$$

$$R_{i,t} = \alpha + \beta_{bank 1} R_{bank 1,t} + \epsilon_{i,t}$$

$$R_{i,t} = \alpha + \beta_{bank 2} R_{bank 2,t} + \epsilon_{i,t}$$

and the average beta coefficients from the following regression model:

$$R_{i,t} = \alpha + \beta_{lead} R_{lead,t} + \beta_{bank 1} R_{bank 1,t} + \beta_{bank 2} R_{bank 2,t} + \epsilon_{i,t}$$

where $R_{i,t}$ is the return on stock i in period t , $R_{lead,t}$ is the return on a value weighted portfolio of all other firms associated with the bank that led the IPO, and $R_{bank 1,2}$ ($R_{bank 2,3}$) is the return on a value weighted portfolio of all firms associated with the bank ranked directly above (below) the bank leading the IPO in terms of underwriting volume in the year of the IPO. A bank is considered to be associated with a firm if it was the sole lead underwriter for the firm’s IPO or seasoned equity offering (SEO) within the last two years. The sample consists of all IPOs that were led by a single underwriter that ranked among the top 25 banks in terms of underwriting volume in the year of the IPO. In addition, we consider only offerings in which the firm did not have another equity offering in the 12 months following the IPO. The regressions are estimated over the one-year period following the IPO. For the average coefficients and average R^2 values for the control bank portfolios, we test for significant differences relative to those for the portfolio of firms associated with the bank leading the offering. *** (***) [] represents significant differences at the 1% (5%) [10%] level. Standard errors are clustered by offering month.

Panel A: Daily returns		Univariate regressions			Multivariate regression		
		Average R^2			Average betas		
Time period	N	Lead bank	Bank 1	Bank 2	Lead bank	Bank1	Bank2
Panel A: Daily returns							
1980–1989	546	0.068	0.061***	0.060***	0.289	0.215***	0.191***
1990–1999	1,647	0.053	0.047***	0.044***	0.326	0.234***	0.212***
2000–2009	347	0.087	0.078***	0.071***	0.339	0.245***	0.186***
1980–2009	2,540	0.061	0.054***	0.051***	0.320	0.231***	0.204***
Panel B: Weekly returns							
1980–1989	546	0.127	0.112***	0.115***	0.369	0.235***	0.267**
1990–1999	1,647	0.102	0.092***	0.086***	0.410	0.278***	0.227***
2000–2009	347	0.147	0.136***	0.122***	0.386	0.305*	0.271*
1980–2009	2,540	0.113	0.102***	0.097***	0.399	0.273***	0.242***

Read: Table 2:

- For daily returns over 1980–2009, the lead-bank adjusted R^2 is 0.061, compared with 0.054 and 0.051 for adjacent control banks.
- In the multivariate regression, the daily lead-bank beta is 0.320, compared with 0.231 and 0.204 for controls.
- Weekly return results are even larger: lead-bank beta is 0.399, compared with 0.273 and 0.242.

Economic message. The same-underwriter network predicts both higher return comovement and greater explanatory power after IPOs.

5.2 Figures 1–4 and Table 3: IPO network dynamics and institutional holdings

Read:

- Figures 1 and 2 show that lead-bank coefficients and adjusted R^2 are larger than control-bank values in the months after the IPO.
- Table 3 shows that the full-sample fraction of the IPO firm held by associated institutions is 0.016 for the IPO bank, versus 0.010 and 0.007 for controls.

- Table 3 also shows that the number of associated institutions is 1.829 for the IPO bank, versus 1.239 and 0.819 for controls.
- Figures 3 and 4 show that these ownership differences persist over the first four quarters after the IPO.

Economic message. The stock-price comovement pattern is matched by investor ownership patterns: the lead bank's investor clientele owns more of the IPO firm.

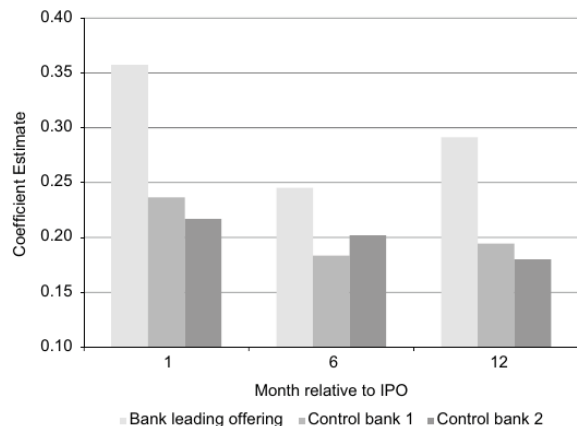


fig. 1. Average coefficients from multivariate regressions. This figure depicts the evolution over time of the estimated coefficients of the following regression model:

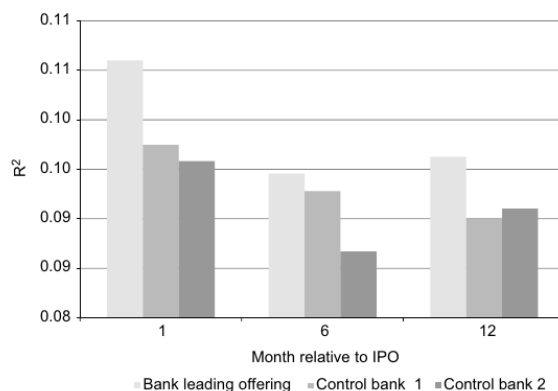


Fig. 2. Average adjusted- R^2 from univariate regressions. This figure depicts the evolution over time of the adjusted- R^2 of the following regression models:

Table 3
Institutional holdings in the first quarter after the initial public offering (IPO).

This table presents average measures of institutional holdings for institutions that are associated with the bank leading the IPO of a given firm as well as the holdings of institutions associated with the two control banks. An institution is considered to be associated with a bank if it ranks above the 75th percentile in terms of the fraction of bank-associated firms that it holds a position in during the quarter immediately following the firm's IPO. We exclude institutions that are associated with more than one of the three banks considered. Panel A presents the fraction of the firm held by institutions, and Panel B presents the total number of such institutions with a position in the stock. For control banks 1 and 2, we test for differences relative to the holdings of the bank leading the IPO. *** (**) represents significance at the 1% (5%) level.

Time Period	N	IPO Bank	Control 1	Control 2
<i>Panel A: Fraction of firm held by associated institutions</i>				
1980–1989	546	0.013	0.007***	0.005***
1990–1999	1647	0.018	0.011***	0.007***
2000–2009	347	0.013	0.010**	0.007***
1980–2009	2540	0.016	0.010***	0.007***
<i>Panel B: Number of associated institutions</i>				
1980–1989	546	1.046	0.641***	0.473***
1990–1999	1647	1.995	1.305***	0.866***
2000–2009	347	2.274	1.867***	1.135***
1980–2009	2540	1.829	1.239***	0.819***

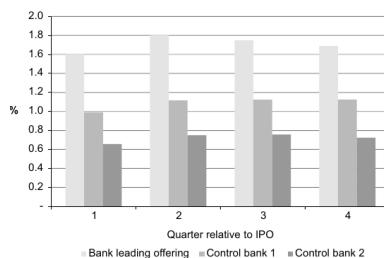


fig. 3. Fraction of firm held by associated institutions. This figure depicts the evolution of the fraction of the initial public offering (IPO) firm held by institutions associated with the lead investment bank and the banks ranked directly above (control bank 1) and below (control bank 2) the lead bank in terms of underwriting volume in the year of the IPO. The sample consists of all IPOs that were led by a single underwriter that ranked among the top 25 banks in terms of underwriting volume in the year of the IPO. In addition, we consider only offerings in which the firm did not have another equity offering in the 12 months following the IPO.

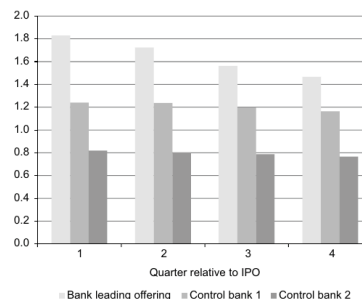


Figure 4. Number of associated institutions. This figure depicts the evolution of the number of institutions holding shares of the initial public offering

5.3 Figure 5 + Tables 4–5: SEO switches and old/new bank comovement

Read: Table 4:

- For daily returns, firms that keep the same bank have a full-sample beta increase of 0.06, while switchers have a beta increase of 0.10.
- The difference test is statistically significant: 2.85 for beta and 3.51 for adjusted R^2 .
- Weekly return results are similar, with switchers showing a 0.10 beta increase and 0.029 increase in adjusted R^2 .

Read: Table 5:

- For daily returns, the new-bank beta rises from 0.39 to 0.46 for the full sample of switchers.
- The old-bank beta stays roughly flat at about 0.34.
- The new-minus-old difference-in-differences is 0.06 for daily returns and 0.11 for weekly returns.

Economic message. When a firm changes underwriters, its return comovement migrates toward the new bank network. This is the paper’s cleanest network evidence.

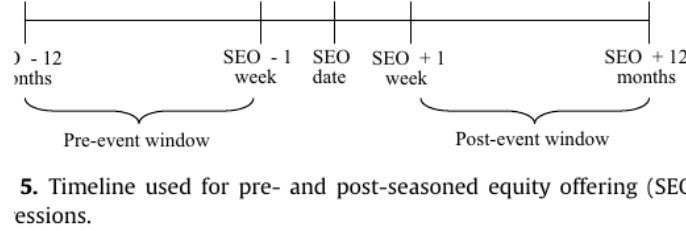


Figure 1: Pre/post SEO event window

Table 4
Comovement with other stocks associated with the bank leading the seasoned equity offering (SEO).
This table reports the estimated coefficients of the following regression model:

$$R_{it} = \alpha_i + \beta_{same} R_{same,t} + \beta_{diff} R_{diff,t} + \epsilon_{it}$$

where R_{it} is the return on stock i in period t , $R_{same,t}$ is the return on a value weighted portfolio of all other firms associated with the bank that led the SEO. A bank is associated with a firm if it was the sole lead underwriter for the firm’s initial public offering (IPO) or SEO in the last two years. The sample is all SEOs led by a single underwriter ranked in the top 25 by underwriting volume in the year of the IPO and that had a single underwriter lead their most recent equity offering. Offerings are divided into two categories: those in which the firm used the same underwriter as its most recent offering and those with a new underwriter. Regressions are estimated for a pre-SEO and post-SEO window of one year before and after the offering, excluding the week before and the week after the offering. “Diff” is the difference between the post-offering coefficient and the pre-offering coefficient. T -tests are performed to determine whether the cross-sectional mean of “Diff” is significantly different from zero. The standard errors for the t -statistics are clustered by offering month, and the t -statistics are reported in parentheses. The columns labeled “Difference” present t -statistics for the tests of the null hypothesis that the means are equal between the switching and nonswitching firms.

	Used same bank for SEO as previous offering							Used different bank for SEO							Difference		
	Average beta			Average R ²				Average Beta			Average R ²						
	N	Pre	Post	Diff	Pre	Post	Diff	N	Pre	Post	Diff	Pre	Post	Diff	Beta	R ²	
Panel A: Daily returns																	
First SEO	655	0.62	0.70	(4.15)	0.062	0.084	(5.36)	550	0.54	0.68	(6.43)	0.050	0.080	(7.21)	2.32	1.77	
Later SEO	856	0.63	0.68	(2.90)	0.04	0.096	(1.76)	808	0.58	0.66	(5.02)	0.08	0.080	(4.60)	0.021	1.87	3.19
Full sample	1,511	0.63	0.69	(4.37)	0.081	0.095	(3.80)	1,358	0.57	0.67	(7.05)	0.068	0.093	(6.41)	0.025	2.85	3.51
Panel B: Weekly Returns																	
First SEO	655	0.77	0.78	(0.43)	0.01	0.114	(1.88)	550	0.62	0.76	(4.25)	0.084	0.124	(6.54)	0.040	3.14	3.58
Later SEO	856	0.71	0.75	(0.44)	0.04	0.144	(1.05)	808	0.64	0.71	(0.77)	0.116	0.138	(2.22)	0.020	1.20	2.12
Full sample	1,511	0.74	0.76	(1.39)	0.131	0.141	(1.71)	1,358	0.63	0.73	(3.25)	0.103	0.132	(3.64)	0.029	3.08	3.81

Table 5
Comovement around seasoned equity offerings (SEOs).
This table reports changes in the slope and the fit of regressions of returns for firms issuing seasoned equity. For each firm which uses a different bank as lead manager for its SEO relative to its previous offering, we regress stock returns on the returns of a value-weighted portfolio of firms associated with the old investment bank and the new investment bank. Firms are considered to be associated with a bank if their IPO or SEO was led by that bank in the last two years. For each offering, we estimate bivariate regressions separately for the one-year period before (pre) and after (post) SEOs as follows:

$$R_{it} = \alpha_i + \beta_{new} R_{new,t} + \beta_{old} R_{old,t} + \epsilon_{it}$$

where R_{it} is the return on stock i in period t , $R_{new,t}$ is the return on a value-weighted portfolio of all firms associated with the new investment bank, and $R_{old,t}$ is the return on a value-weighted portfolio of all firms associated with the old investment bank. Firm i is eliminated from these portfolios, and we exclude the week before and after the SEO. “Diff” is the difference between the post-offering coefficient and the pre-offering coefficient. T -tests are performed to determine whether the cross-sectional mean of “Diff” is significantly different from zero. The standard errors for the t -statistics are clustered by offering month and the t -statistics are reported in parentheses. Panel A shows the results for daily returns, and Panel B shows the results for weekly returns.

	N	New bank – average beta			Old bank – average beta			Diff (new) - Diff (old)
		Pre	Post	Diff	Pre	Post	Diff	
Panel A: Daily returns								
First SEO	393	0.39	0.50	0.11	0.33	0.32	0.00	0.11
		(4.83)		(4.83)			(-0.08)	(3.41)
Later SEO	647	0.39	0.44	0.05	0.34	0.35	0.01	0.04
		(3.27)		(3.27)			(0.66)	(1.73)
Full sample	1,040	0.39	0.46	0.07	0.34	0.34	0.01	0.06
		(5.66)		(5.66)			(0.46)	(3.85)
Panel B: Weekly returns								
First SEO	393	0.44	0.55	0.11	0.37	0.35	-0.02	0.12
		(2.53)		(2.53)			(-0.42)	(1.76)
Later SEO	647	0.42	0.48	0.06	0.38	0.34	-0.04	0.11
		(2.38)		(2.38)			(-1.85)	(2.50)
Full sample	1,040	0.43	0.51	0.08	0.38	0.34	-0.03	0.11
		(3.38)		(3.38)			(-1.57)	(2.98)

5.4 Table 6 + Tables 7–8: forced switches and institutional ownership changes

Read: Table 6:

- In the bank merger/closure sample, the full-sample daily beta increase is 0.197, with $t = 3.64$.
- The full-sample daily adjusted R^2 increase is 0.042, with $t = 3.56$.
- The results remain positive using weekly returns.

Read: Tables 7–8:

- Around SEOs, the fraction held by associated institutions rises for both non-switchers and switchers, but the increase is larger for switchers.
- For switchers, new-bank institutions increase ownership more than old-bank institutions.
- In Table 8, the full-sample new-bank institution count rises by 1.34, while the old-bank institution count rises by only 0.49; the difference is 0.84.

Economic message. Ownership changes follow the same direction as return comovement changes: moving to a new bank also moves the investor clientele.

Table 6

Comovement with other stocks associated with the bank leading seasoned equity offering SEO – bank mergers and closures.

This table reports the estimated coefficients and adjusted R^2 values of the following regression model:

$$R_{i,t} = \alpha_1 + \beta_{Lead,t} R_{Lead,t} + e_{i,t},$$

where $R_{i,t}$ is the return on stock i in period t , $R_{Lead,t}$ is the return on a value weighted portfolio of all firms associated with the bank that led the SEO. A bank is considered to be associated with a firm if it was the sole lead underwriter for the firm's initial public offering (IPO) or SEO within the last two years. The sample consists of all SEOs that were led by a single underwriter that ranked among the top 25 banks in terms of underwriting volume in the year of the IPO and had a single underwriter lead their most recent equity offering. This table reports results for a sample of firms that switched banks due to a merger of their former lead bank or the closure of their former lead bank. The regressions are estimated for a pre-SEO and post-SEO window of one year before and after the offering, excluding the week before and the week after the offering. "Diff" is the difference between the post-offering coefficient and the pre-offering coefficient. T -tests are performed to determine whether the cross-sectional mean of "Diff" is significantly different from zero. The standard errors for the t -statistics are clustered by offering month and the t -statistics are reported in parentheses.

	N	Average beta			Average R ²		
		Pre	Post	Diff	Pre	Post	Diff
<i>Panel A: Daily returns</i>							
First SEO	20	0.507	0.778	0.271 (2.67)	0.066	0.096	0.030 (1.83)
Later SEO	26	0.636	0.776	0.139 (2.46)	0.075	0.126	0.051 (3.08)
Full sample	46	0.580	0.777	0.197 (3.64)	0.071	0.113	0.042 (3.56)
<i>Panel B: Weekly returns</i>							
First SEO	20	0.600	0.894	0.294 (1.72)	0.090	0.145	0.055 (1.67)
Later SEO	26	0.717	0.802	0.086 (0.64)	0.123	0.167	0.044 (1.41)
Full sample	46	0.666	0.842	0.176 (1.65)	0.108	0.157	0.049 (2.14)

Table 7

Changes in institutional holdings around the seasoned equity offering (SEO).

This table presents average measures of institutional holdings for institutions associated with the bank leading a firm's SEO. An institution is considered to be associated with a bank if it ranks above the 75th percentile in terms of the fraction of bank-associated firms that it holds a position in during the quarter immediately following those bank-associated firms' initial public offerings. Panel A presents the fraction of the firm held by institutions, and Panel B presents the total number of such institutions with a position in the stock. The pre-event observation comes from the period (–180 days, –90 days) relative to the offering, and the post-event observation comes from the period (0, +90 days) relative to the offering. "Diff" is the difference between the post-offering coefficient and the pre-offering coefficient. T -tests are performed to determine whether the cross-sectional mean of "Diff" is significantly different from zero. The standard errors for the t -statistics are clustered by offering month and the t -statistics are reported in parentheses. The columns labeled "Difference" present t -statistics for the tests of the null hypothesis that the means of the switching and nonswitching firms are equal.

	Used same bank for SEO				Used different bank for SEO				Difference
	N	Pre	Post	Diff	N	Pre	Post	Diff	
Panel A: Fraction of firm held by associated institutions									
First SEO	655	0.157	0.215	0.058 (10.73)	550	0.100	0.166	0.066 (13.89)	1.34
Later SEO	856	0.172	0.189	0.018 (5.58)	808	0.152	0.183	0.031 (8.03)	3.41
Full sample	1,511	0.165	0.200	0.035 (9.49)	1358	0.131	0.176	0.045 (12.84)	2.92
Panel B: Number of associated institutions with a stake in firm									
First SEO	655	13.702	17.056	3.354 (9.14)	550	9.022	13.502	4.480 (15.31)	3.27
Later SEO	856	17.731	19.362	1.631 (6.33)	808	16.978	19.780	2.802 (8.35)	3.91
Full sample	1,511	15.985	18.363	2.378 (8.81)	1,358	13.756	17.237	3.482 (12.39)	4.85

5.5 Table 9: ownership changes explain comovement changes

Read:

- Table 9 regresses changes in beta and adjusted R^2 on ownership changes and offering characteristics.
- The different-bank dummy is positive in both daily and weekly panels.
- Changes in the fraction held by lead-bank institutions and changes in the number of lead-bank institutions are positively related to changes in comovement.
- In the daily panel, *Diff. Sum Inst.* has a large positive coefficient for both $\Delta\beta$ and ΔR^2 .

Economic message. The mechanism is not only that firms switch banks. The comovement change is stronger when ownership by the lead bank's investor network changes more.

5.6 Figure 6 + Tables 10–11: trading activity and robustness

Read: Figure 6:

- Before the SEO, the old underwriter has a much larger Nasdaq dealer market share than the future new underwriter.
- Around the event date, the roles reverse: the new underwriter's market share jumps above 20%, while the old underwriter's share falls toward 10%.

Table 8

Changes in institutional ownership around the seasoned equity offering (SEO) when firms switch banks.

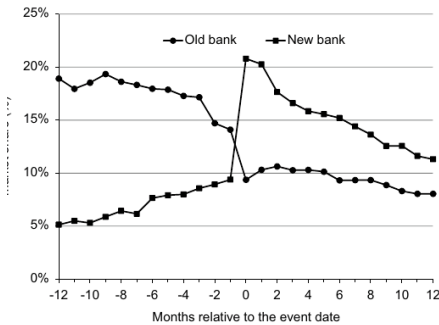
This table presents the institutional ownership characteristics of firms around the SEO when firms switch banks. Panel A shows the total percentages of the firm held by old and new bank institutions. Panel B shows the number of institutions associated with the old and new banks that hold a stake in the firm. The pre-event observation comes from the period (−180 days, −90 days) relative to the offering, and the post-event observation comes from the period (0, +90 days) relative to the offering. An institution is considered to be associated with a bank if it ranks above the 75th percentile in terms of the fraction of bank-associated firms that it holds a position in during the quarter immediately following those bank-associated firms' initial public offering. We exclude institutions associated with both the old and new bank. "Diff" is the difference between the post-offering coefficient and the pre-offering coefficient. T-tests are performed to determine whether the cross-sectional mean of "Diff" is significantly different from zero. The standard errors for the t-statistics are clustered by offering month and the t-statistics are reported in parentheses.

	N	New bank institutions			Old bank institutions			
		Pre	Post	Diff	Pre	Post	Diff	Diff-in-Diff
Panel A: Total fraction held by associated institutions								
First SEO	393	0.035	0.064	0.030 (8.00)	0.027	0.034	0.006 (3.58)	0.023 (5.38)
Later SEO	647	0.046	0.056	0.010 (4.71)	0.036	0.042	0.005 (3.41)	0.004 (1.81)
Full sample	1,040	0.042	0.059	0.017 (8.03)	0.033	0.039	0.006 (4.58)	0.012 (4.77)
Panel B: Total number of associated institutions								
First SEO	393	3.25	5.20	1.95 (11.52)	2.74	3.35	0.61 (4.78)	1.33 (6.01)
Later SEO	647	5.54	6.50	0.96 (5.72)	4.65	5.07	0.42 (3.27)	0.54 (3.40)
Full sample	1,040	4.67	6.01	1.34 (9.52)	3.93	4.42	0.49 (4.66)	0.84 (5.93)

Read: Tables 10–11:

- Table 10 shows that turnover comovement increases for firms that switch underwriters, especially using weekly turnover.
- Table 11 shows that switchers exhibit larger new-bank comovement changes than matched non-switchers.
- For the full daily sample in Table 11, the difference-in-differences is 0.056; for weekly returns it is 0.138.

Economic message. Network migration is visible in trading and survives matched-sample tests, making it harder to attribute the result to simple firm-level risk changes.



6. Time series of market-making activity. This figure presents the series of Nasdaq dealer market shares in event time around the th of the seasoned equity offering (SEO) for a sample of firms that shed underwriters from their initial public offering (IPO) to their SEO.

Table 9

Relation between change in comovement and change in institutional ownership.

This table presents regression results relating changes in beta and adjusted R^2 from the regressions in Table 4 to changes in institutional holdings by institutions associated with the lead bank, and other firm or offering characteristics. *Different Bank* is a dummy equal to one if the firm switched banks for this offering. *First SEO Dummy* is equal to one if the offering is the first seasoned offering (SEO) after its initial public offering (IPO). *Relative Size of SEO* is the ratio of the offering size to market capitalization prior to the offering. *Years Since Last Offering* is the number of years since the firm's prior equity offering. *Bank Top 10* is a dummy variable equal to one if the lead bank was in the top ten banks by underwriting volume the year of the offering. *New Coverage* is a dummy equal to one if the lead bank initiated research coverage around the offering. *Diff. Sum Inst.* is the change in the fraction of the firm held by institutions associated with the lead bank. *Diff. Sum All Inst.* is the change in the fraction of the firm held by all institutions. *Diff. Num. Inst.* is the change in the number of institutions associated with the lead bank. *Diff. Num. All Inst.* is the change in the total number of institutions. An institution is considered associated with a bank if it ranks above the 75th percentile by the fraction of bank-associated firms that it holds around the bank-associated firms' IPOs. Panel A presents the results for the coefficient estimates from the daily return regressions, and Panel B presents the results for the coefficient estimates from the weekly return regressions. *, **, and *** represent significance at the 10%, 5%, and 1% level, respectively.

	$\Delta\beta$			ΔR^2		
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Daily returns						
Constant	3.492*	3.023	2.343	0.262	0.219	−0.714
	(1.713)	(1.467)	(1.103)	(0.589)	(0.489)	(−1.570)
<i>Different bank</i>	3.671**	3.324**	2.933*	0.861**	0.779**	0.647*
	(2.294)	(2.092)	(1.856)	(2.466)	(2.249)	(1.914)
<i>First SEO dummy</i>	4.072***	1.872	2.614*	1.227***	0.753**	0.761**
	(2.666)	(1.167)	(1.687)	(3.599)	(2.160)	(2.295)
<i>Relative size of SEO</i>	0.013**	0.005	0.008	0.001	−0.001	−0.000
	(2.343)	(0.856)	(1.489)	(0.702)	(−0.763)	(−0.167)
<i>Years since last offering</i>	0.391	0.410	0.333	0.167***	0.171**	0.130**
	(1.448)	(1.530)	(1.246)	(2.836)	(2.929)	(2.272)
<i>Bank Top 10</i>	−2.363	−2.787*	−3.438**	0.278	0.155	−0.140
	(−1.438)	(−1.702)	(−2.113)	(0.777)	(0.425)	(−0.462)
<i>New coverage</i>	−2.459	−3.451*	−3.875**	−0.250	−0.482	−0.715*
	(−1.280)	(−1.805)	(−2.038)	(−0.508)	(−1.156)	(−1.759)
<i>Diff. Sum Inst.</i>		40.774***			30.988***	
		(3.895)			(4.817)	
<i>Diff. Sum All Inst.</i>		14.761**			2.236	
		(2.015)			(1.401)	
<i>Diff. Num. Inst.</i>			0.989***			0.252***
			(7.511)			(8.933)
<i>Diff. Num. All Inst.</i>				0.069		0.067**
				(1.359)		(6.181)
<i>N</i>	2,759	2,759	2,759	2,759	2,759	2,759
<i>R</i> ²	0.011	0.027	0.039	0.012	0.030	0.077
Panel B: Weekly returns						
Constant	2.782	2.097	2.082	0.633	0.574	−0.561
	(0.858)	(0.637)	(0.610)	(0.868)	(0.775)	(−0.738)
<i>Different bank</i>	7.365***	7.038***	6.715**	1.801**	1.821**	1.679**
	(2.893)	(2.770)	(2.644)	(3.318)	(3.185)	(2.969)
<i>First SEO dummy</i>	1.039	−1.210	−0.225	0.898	0.424	0.403
	(0.418)	(−0.472)	(−0.080)	(1.686)	(0.735)	(0.726)
<i>Relative size of SEO</i>	0.011	0.003	0.006	0.002	0.000	0.001
	(1.202)	(0.270)	(0.703)	(0.809)	(−0.089)	(−0.312)
<i>Years since last offering</i>	−0.111	−0.093	−0.154	−0.019	−0.015	−0.064
	(−0.262)	(−0.217)	(−0.358)	(−0.199)	(−0.138)	(−0.666)
<i>Bank Top 10</i>	−2.553	−2.874	−3.449	−0.340	−0.455	−0.804
	(−0.977)	(−1.098)	(−1.320)	(−0.579)	(−0.773)	(−1.382)
<i>New coverage</i>	−1.175	−2.136	−2.397	0.267	0.039	−0.231
	(−0.385)	(−0.696)	(−0.785)	(0.387)	(0.057)	(−0.339)
<i>Diff. Sum Inst.</i>		33.789**			10.411**	
		(2.019)			(2.785)	
<i>Diff. Sum All Inst.</i>		18.370			2.507	
		(1.568)			(0.951)	
<i>Diff. Num. Inst.</i>			0.889***			0.251***
			(4.201)			(5.321)
<i>Diff. Num. All Inst.</i>			0.038			0.083**
			(0.463)			(4.570)
<i>N</i>	2,759	2,759	2,759	2,759	2,759	2,759
<i>R</i> ²	0.005	0.011	0.013	0.007	0.013	0.036

Table 10

Turnover comovement with other stocks associated with same bank around the seasoned equity offering (SEO).

This table reports the estimated coefficients of the following regression model:

$$\Delta TO_{i,t} = \alpha_i + \beta_{cont,i} \Delta TO_{cont,t} + \epsilon_{i,t}$$

where $\Delta TO_{i,t}$ is the percentage change in turnover on stock i in period t , $\Delta TO_{cont,t}$ is the percentage change in turnover on a value weighted portfolio of all firms associated with the bank that led the SEO. A bank is considered to be associated with a firm if it was the sole lead underwriter for the firm's IPO/SEO within the last two years. The sample is all SEOs led by a single underwriter ranked in the top 25 by underwriting volume and that had a single underwriter lead their most recent equity offering. Offerings are divided into those in which the firm used the same underwriter as in its previous offering and those that used a new underwriter. Regressions are estimated for a pre-SEO and post-SEO window of one year excluding the week before and after the offering. "Diff" is the difference between the post-offering coefficient and the pre-offering coefficient. The standard errors for the t-statistics are clustered by offering month and the t-statistics are reported in parentheses. The columns labeled "Difference" present t-statistics for the tests of the null hypothesis that the means for the switching and non-switching firms are equal.

	N	Used same bank for SEO as previous offering						Used different bank for SEO						Difference		
		Average beta			Average R ²			Average beta			Average R ²					
		Pre	Post	Diff	Pre	Post	Diff	N	Pre	Post	Diff	Pre	Post	Diff	Beta	R ²
Panel A: Daily turnover																
First SEO	655	0.43	0.47	0.04 (1.76)	0.028	0.028	−0.001 (−0.19)	550	0.40	0.41	0.02 (0.92)	0.024	0.024	0.000 (0.01)	−0.80	0.22
Later SEO	856	0.47	0.45	−0.02 (−1.47)	0.038	0.033	−0.004 (−0.98)	808	0.40	0.42	0.02 (1.21)	0.029	0.033	0.003 (1.50)	2.29	2.43
Full sample	1,511	0.45	0.46	0.00 (0.34)	0.033	0.031	−0.003 (−0.72)	1,358	0.40	0.42	0.02 (1.36)	0.027	0.029	0.002 (0.92)	0.88	1.94
Panel B: Weekly turnover																
First SEO	655	0.78	0.79	0.02 (0.52)	0.116	0.132	0.016 (2.31)	550	0.63	0.77	0.14 (4.44)	0.086	0.129	0.043 (6.54)	3.23	3.37
Later SEO	856	0.73	0.77	0.04 (1.94)	0.147	0.160	0.012 (1.52)	808	0.65	0.73	0.08 (3.62)	0.118	0.145	0.027 (4.09)	1.35	2.01
Full sample	1,511	0.75	0.78	0.03 (1.64)	0.134	0.148	0.014 (2.18)	1,358	0.64	0.75	0.11 (5.15)	0.105	0.138	0.033 (6.07)	3.22	3.60

Table 11

Matched sample analysis.

This table reports changes in the coefficient estimates from regressions of returns for a firm that switches banks for its seasoned equity offering (SEO) relative to a firm that does not. For each firm that uses a different bank as lead manager of its SEO relative to the previous offering, we find a matching firm that completed an SEO within 60 days of the firm's SEO but did not switch banks. The match is made based on the relative size of the offering and the time since the last equity offering. We regress stock returns on the returns of a value-weighted portfolio of firms associated with the old investment bank and the new investment bank. Firms are considered to be associated with a bank if that bank led their initial public offering (IPO) or SEO in the last two years. For each offering, we estimate bivariate regressions separately for the one-year period before and after SEOs as follows:

$$R_{i,t} = \alpha_i + \beta_{New,i} R_{New,i,t} + \beta_{Old,i} R_{Old,i,t} + \varepsilon_{i,t},$$

where $R_{i,t}$ is the return on stock i in period t , $R_{New,i,t}$ is the return on a value-weighted portfolio of all firms associated with the new investment bank, and $R_{Old,i,t}$ is the return on a value-weighted portfolio of all firms associated with the old investment bank. Firm i is eliminated from these portfolios, and we exclude the week before and after the SEO. We also run these regressions for the matching firms and report averages of the difference between the change in coefficients for the firm of interest and the matching firm. Standard errors are clustered by month. T -statistics are reported in parentheses. Panel A shows the results for daily returns, and Panel B shows the results for weekly returns.

	$\Delta\beta_{New} - \Delta\beta_{New,Match}$	$\Delta\beta_{Old} - \Delta\beta_{Old,Match}$	Diff-in-diff.
<i>Panel A: Daily returns</i>			
First SEO	0.092 (4.05)	-0.012 -(0.60)	0.103 (3.29)
Later SEO	0.035 (2.20)	0.008 (0.48)	0.027 (1.12)
Full sample	0.057 (4.21)	0.001 (0.04)	0.056 (3.05)

6 Short summary

- The paper studies whether investment banks create segmented investor networks that affect stock prices and trading.
- It defines a bank network by recent underwriting relationships: firms are associated with a bank if that bank led their IPO or SEO within the last two years.
- IPO firms comove more with stocks in the same lead-underwriter network than with stocks associated with adjacent control banks.
- Institutional investors associated with the lead bank hold larger stakes in the IPO firm than institutions associated with adjacent banks.
- When firms switch underwriters at SEOs, they begin to comove more with the new bank's portfolio and not with the old bank's portfolio.
- Forced-switch samples caused by bank mergers and closures show similar results.
- Institutional ownership changes are positively related to comovement changes.
- Nasdaq market-making activity also moves from the old underwriter to the new underwriter around the SEO.
- Turnover comovement and matched-sample tests support the interpretation that the effect comes from investor networks rather than pure risk factors.
- The paper's conclusion is that underwriting relationships can segment capital markets and affect asset prices.

7 Conclusion

7.1 Core conclusion

Grullon, Underwood, and Weston show that investment banking networks affect stock returns and trading behavior. Firms associated with the same lead underwriter comove more than comparable firms associated with nearby banks. When a firm switches underwriters, its comovement shifts toward the new bank and away from the old bank.

7.2 Economic intuition

Investment banks repeatedly market securities to investor clienteles, provide information about issuers, and maintain trading and market-making relationships. These activities create segmented groups of investors who pay attention to, own, and trade stocks connected to their preferred bank. As a result, shocks to investor demand or information inside one bank network can create comovement across stocks in that network.

7.3 Takeaway

The paper extends the literature on comovement beyond indexes, styles, geography, and analyst coverage. It shows that corporate finance relationships can become asset-pricing links. The underwriter is not only a certifier at issuance; it is also an information and investor-network hub that can shape the firm's secondary-market behavior.